

SOC Internship – Day 27 Notes

EDR and XDR Monitoring

Objective

Understand how Security Operations Center (SOC) analysts use EDR and XDR tools to detect, investigate, and respond to cyber threats.

1. What is EDR?

EDR (Endpoint Detection and Response) is a cybersecurity solution that continuously monitors endpoint devices such as:

- Computers
- Laptops
- Servers
- Mobile devices

It helps detect suspicious activities and respond to threats quickly.

Main Functions of EDR

- Monitor endpoint behavior
- Detect malware and attacks
- Collect security logs
- Alert SOC analysts
- Support incident response

Examples of Endpoint Activities

- Unauthorized file changes
- Suspicious PowerShell commands
- Unknown application execution
- Credential dumping attempts

2. What is XDR?

XDR (Extended Detection and Response) is an advanced security solution that combines multiple security layers into one platform.

XDR Collects Data From

- Endpoints
- Email systems
- Cloud platforms

- Networks
- Firewalls
- Identity systems

Purpose of XDR

- Correlate alerts from multiple sources
- Improve threat visibility
- Reduce false positives
- Provide centralized monitoring

3. Difference Between EDR and XDR

Feature	EDR	XDR
Full Form	Endpoint Detection and Response	Extended Detection and Response
Focus Area	Endpoint devices only	Multiple security layers
Visibility	Limited to endpoints	Organization-wide
Data Sources	Endpoint telemetry	Endpoint, cloud, email, network
Detection Capability	Endpoint threats	Advanced correlated attacks
SOC Usage	Endpoint investigation	Unified threat detection

4. Common Threats Detected

EDR and XDR platforms help identify different types of cyber threats such as:

Malware Execution

Malicious software running on a system.

Ransomware Behavior

Encryption of files and suspicious mass file modifications.

Credential Theft

Attackers stealing usernames and passwords.

Suspicious PowerShell Activity

Encoded or hidden PowerShell commands used by attackers.

Lateral Movement

Attackers moving from one system to another inside the network.

5. Example Investigation Scenario

Alert Triggered

Encoded PowerShell execution detected

Findings

- Suspicious process activity
- External malicious network connection

Analysis

This behavior may indicate:

- Malware infection
- Command-and-control communication
- Remote attacker activity

SOC Analyst Actions

1. Investigate the endpoint
2. Check process tree
3. Analyze network connections
4. Isolate infected device
5. Remove malicious files
6. Report incident

6. Detection Techniques Used in EDR/XDR

Behavioral Analysis

Detect abnormal behavior patterns.

IOC Matching

Compare activities with known Indicators of Compromise (IOCs).

Process Monitoring

Track running processes and parent-child relationships.

File Activity Analysis

Monitor suspicious file creation, deletion, or modification.

7. SOC Workflow

Endpoint Activity

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EDR/XDR Detection



Alert Triggered



SOC Investigation

8. SOC Roles in EDR/XDR Monitoring

L1 Analyst

- Monitor alerts
- Perform initial triage
- Escalate serious incidents

L2 Analyst

- Investigate suspicious activities
- Analyze malware behavior
- Perform deeper incident analysis

L3 Analyst

- Improve detection rules
- Develop threat hunting queries
- Enhance SOC detection logic

Assignment Answers

1. Explain EDR and XDR

EDR monitors endpoint devices and detects suspicious activities for faster incident response.

XDR extends detection capabilities by integrating endpoint, network, cloud, and email telemetry into one centralized platform.

EDR

Focuses on endpoints
Limited visibility
Detects endpoint threats
Single-source telemetry

XDR

Covers multiple security domains
Centralized visibility
Detects correlated advanced attacks
Multi-source telemetry

3. Malware Detection Scenario Analysis

An alert shows encoded PowerShell execution. The SOC analyst investigates and discovers suspicious processes and external malicious connections. This indicates a possible malware infection. The analyst isolates the affected endpoint, analyzes logs, removes the malware, and reports the incident.